

SLAI MF2601: Homework I

Name: _____ Student ID: _____

Due: September 30, 2026

General instructions. Unless a problem says otherwise, all matrices are real; use MATLAB or Python for numerical work. Use L^AT_EX to typeset the homework and compile to PDF; you may modify the source code provided. Submit a single zip file containing the PDF (with the theory and numerical-experiment results) together with a MATLAB or Python code file, to the TA's email with subject line **NLA HW1**.

[Do not submit this part] Problem 0: Preliminary questions

This is a study aid only. Do not submit answers to these questions. You may use any resources, including AI tools, to answer these questions to ensure a **basic** understanding of linear algebra.

What is the definition of **linear space** or vector space?

What is the definition of a **basis** of a linear space?

What is the definition of the **dimension** of a linear space?

What is the definition of **linear mapping**?

What is the definition of the **image and kernel** of a linear mapping?

What is the **relationship between matrix and linear mapping** under a fixed basis?

What happens to the matrix of a linear mapping if we change the basis?

What happens to their matrices if two linear mappings are composed?

What is the **rank** and **nullity** of a matrix?

Kernel – nullity. Image – rank. What are their relationships?

What is the **rank-nullity** theorem?

What is the definition of a linear **subspace** and the summation and intersection between two subspaces?

How do you prove the dimension formula for the sum of two subspaces?

$$\dim(S + T) = \dim S + \dim T - \dim(S \cap T)$$

What is the definition of inner product?

What is the definition of orthogonal vectors?

Problem 1

Page 16, Problem 2.6. If u and v are m -vectors, the matrix $A = I + uv^H$ is known as a *rank-one perturbation of the identity*. Show that if A is nonsingular, then its inverse has the form $A^{-1} = I + \alpha uv^H$ for some scalar α , and give an expression for α . For what u and v is A singular? If it is singular, what is $\text{null}(A)$?

Problem 2

Page 24, Problem 3.6. Let $\|\cdot\|$ denote any norm on $\mathbb{C}^m$. The corresponding *dual norm* $\|\cdot\|'$ is defined by the formula

$$\|x\|' = \sup_{\|y\|=1} |y^H x|.$$

- (a) Prove that $\|\cdot\|'$ is a norm.
- (b) Let $x, y \in \mathbb{C}^m$ with $\|x\| = \|y\| = 1$ be given. Show that there exists a rank-one matrix $B = yz^H$ such that $Bx = y$ and $\|B\| = 1$, where $\|B\|$ is the matrix norm of B induced by the vector norm $\|\cdot\|$. You may use the following lemma, without proof: given $x \in \mathbb{C}^m$, there exists a nonzero $z \in \mathbb{C}^m$ such that $|z^H x| = \|z\|' \|x\|$.

Problem 3

Page 31, Problem 4.4. Two matrices $A, B \in \mathbb{C}^{m \times m}$ are *unitarily equivalent* if $A = QBQ^H$ for some unitary $Q \in \mathbb{C}^{m \times m}$. Is it true or false that A and B are unitarily equivalent if and only if they have the same multiset of singular values, counted with multiplicity? *If yes, please give the proof; otherwise provide an example.*

Problem 4

Page 37, Problem 5.4. Suppose $A \in \mathbb{C}^{m \times m}$ has an SVD $A = U\Sigma V^H$. Find an eigenvalue decomposition (5.1) of the $2m \times 2m$ Hermitian matrix

$$\begin{bmatrix} 0 & A^H \\ A & 0 \end{bmatrix}.$$

Problem 5. SVD application: orthogonal Procrustes

Suppose the rows of $A, B \in \mathbb{R}^{m \times n}$ are corresponding observations in two coordinate systems. The orthogonal Procrustes problem is

$$\min_{Q^T Q = I} \|AQ - B\|_F. \quad (\text{A.1})$$

- A1.** Use the identity $\|M\|_F^2 = \text{tr}(M^T M)$ to expand the squared Frobenius norm in (A.1), discard the Q -independent terms, and show that the problem is equivalent to

$$\max_{Q^T Q = I} \text{tr}(Q^T A^T B). \quad (\text{A.2})$$

A2. Let $A^\top B = U\Sigma V^\top$ be a full SVD. Prove that

$$Q_\star = UV^\top \quad (\text{A.3})$$

is a maximizer of (A.2). Your proof should introduce an orthogonal matrix W and justify the trace inequality used to bound $\text{tr}(\Sigma W)$.

A3. Make the equality case explicit. If $r = \text{rank}(A^\top B)$, show that all maximizers have the form

$$Q = U \begin{bmatrix} I_r & 0 \\ 0 & W_0 \end{bmatrix} V^\top, \quad (\text{A.4})$$

where $O(n-r)$ denotes the set of all $(n-r) \times (n-r)$ orthogonal matrices, i.e., $W_0 \in \mathbb{R}^{(n-r) \times (n-r)}$ with $W_0^\top W_0 = I$. When $r = n$ the block W_0 is absent (0×0) and the formula reduces to $Q = UV^\top$. Conclude precisely when the minimizer is unique.

A4. Numerical experiment. Choose $m = 50$, $n = 5$ (any $m \geq n$ works). Generate a random $A \in \mathbb{R}^{m \times n}$ with i.i.d. $\mathcal{N}(0, 1)$ entries, an orthogonal $Q_0 \in \mathbb{R}^{n \times n}$ (e.g., the Q factor of the QR factorization of a random Gaussian matrix), and a noise matrix $E \in \mathbb{R}^{m \times n}$ with i.i.d. $\mathcal{N}(0, 1)$ entries. For each $\sigma \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$:

- (a) Set $B = AQ_0 + \sigma E$.
- (b) Compute $Q_\star = UV^\top$ from an SVD $A^\top B = U\Sigma V^\top$.
- (c) Report three Frobenius-norm residuals:

$$\epsilon_{\text{orth}} := \|Q_\star^\top Q_\star - I\|_F, \quad \epsilon_{\text{data}} := \|AQ_\star - B\|_F, \quad \epsilon_{\text{align}} := \|Q_\star - Q_0\|_F.$$

Note. The $\sigma = 0$ case is the noise-free sanity check $B = AQ_0$; there $Q_\star$ should recover Q_0 to machine precision and both $\epsilon_{\text{data}}, \epsilon_{\text{align}}$ should be ≈ 0 .

Problem 6. SVD application: canonical correlation analysis

Motivation. Suppose m samples carry measurements from two blocks of variables: $X \in \mathbb{R}^{m \times p}$ (say, p psychological test scores) and $Y \in \mathbb{R}^{m \times q}$ (say, q physical measurements), with rows = samples, columns = variables (so the j -th column $X_j \in \mathbb{R}^m$ holds the m observations of variable j), both blocks *column-centered* (i.e., $\frac{1}{m} \sum_{i=1}^m X_{ij} = 0$ for every column j). The ordinary Pearson correlation $\rho(X_j, Y_k)$ measures the linear association of a *single* pair of variables, but in practice variables within each block are correlated with one another, and any block-level relationship is often split across many variables rather than concentrated in one pair. **Canonical correlation analysis (CCA)** asks for the strongest such *block-level* linear relationship: find linear combinations Xa, Yb (with $a \in \mathbb{R}^p$, $b \in \mathbb{R}^q$) that are maximally correlated. Up to $\min(p, q)$ such pairs (a_i, b_i) are produced, with successive pairs constrained to be orthogonal to the previous ones so that each new pair captures fresh block-level structure. The numbers $\rho_i = \text{Corr}(Xa_i, Yb_i)$ are the *canonical correlations*, the vectors a_i, b_i are *canonical directions*, and the projections Xa_i, Yb_i are *canonical variates*.

The Pearson correlation of two m -vectors u, v (zero-mean, as guaranteed by column-centering) is $\text{Corr}(u, v) = u^\top v / (\|u\|_2 \|v\|_2)$, so the single-pair CCA problem is

$$\max_{a \in \mathbb{R}^p, b \in \mathbb{R}^q} \text{Corr}(Xa, Yb) = \max_{a, b} \frac{a^\top (X^\top Y) b}{\|Xa\|_2 \|Yb\|_2}. \quad (\text{B.1})$$

Each remaining pair (a_i, b_i) , $i \geq 2$, maximizes (B.1) subject to the additional constraints $\text{Corr}(Xa_i, Xa_j) = \text{Corr}(Yb_i, Yb_j) = 0$ for every $j < i$.

Define the sample covariance matrices

$$S_{xx} = \frac{1}{m-1} X^\top X, \quad S_{yy} = \frac{1}{m-1} Y^\top Y, \quad S_{xy} = \frac{1}{m-1} X^\top Y.$$

We assume throughout that S_{xx} and S_{yy} are positive definite, and let $S_{xx}^{1/2}$ and $S_{yy}^{1/2}$ denote their symmetric positive-definite square roots. Whitening by these square roots turns the problem into one SVD.

- B1.** Derive a clean algebraic formulation of the single-pair problem (B.1) using S_{xx} , S_{yy} , S_{xy} .
- B2.** Solve the single-pair formulation from B1. Give closed-form expressions for the first canonical pair (a_1, b_1) and the first canonical correlation ρ_1 .
- B3.** Generalize to all $k = \min(p, q)$ pairs. Give closed-form expressions for every (a_i, b_i) and every ρ_i , and verify $a_i^\top S_{xx} a_j = \delta_{ij}$ and $b_i^\top S_{yy} b_j = \delta_{ij}$.

Problem 7. Lecture 04 supplement

See also: the Lecture 04 supplement at <http://47.107.139.91:2601/files/lecture/lec04.pdf>.

C.1 Max–min characterization (self-contained lemma)

Let $A \in \mathbb{R}^{m \times n}$ with $m \geq n$ and singular values $\sigma_1(A) \geq \dots \geq \sigma_n(A) \geq 0$. For $1 \leq i \leq n$, define

$$\phi_i(A) = \max_{\substack{S \subseteq \mathbb{R}^n \\ \dim S = i}} \min_{\substack{x \in S \\ \|x\|_2 = 1}} \|Ax\|_2.$$

Task C1. Prove that $\phi_i(A) = \sigma_i(A)$.

C.2 Perturbation bound for singular values

Theorem 1 (Weyl-type singular-value bound). For $A, B \in \mathbb{R}^{m \times n}$, $m \geq n$,

$$\max_{1 \leq i \leq n} |\sigma_i(A) - \sigma_i(B)| \leq \|A - B\|_2. \quad (\text{C.1})$$

Task C2. Prove Theorem 1. You may use the max–min characterization $\phi_i(A) = \sigma_i(A)$ from Part C.1. As a sanity check, verify (C.1) numerically on at least five random matrix pairs and report the largest observed ratio $\max_i |\sigma_i(A) - \sigma_i(B)| / \|A - B\|_2$.

C.3 Best rank- k approximation and distance to rank deficiency

Let $A \in \mathbb{R}^{n \times n}$ be nonsingular with SVD $A = U \text{diag}(\sigma_1, \dots, \sigma_n) V^\top$, where $\sigma_1 \geq \dots \geq \sigma_n > 0$.

Task C3. Fix $0 \leq k < n$. Determine the minimum-norm perturbation that makes A rank-deficient, taken over all $E \in \mathbb{R}^{n \times n}$:

$$\min_{\substack{\text{rank}(A+E) \leq k \\ E \in \mathbb{R}^{n \times n}}} \|E\|_2. \quad (\text{C.2})$$